

# okDriver CCTV Surveillance & Video Analytics Platform

*System Architecture & Enterprise Scalability Document (80,000+ Cameras)*

**Target Scale: Prototype: 48 Cameras | Production Deployment: 80,000+ Cameras**

## 1. Multi-Tier Distributed Processing Architecture

Scaling to 80,000 cameras requires a 3-Tier Compute Model (Edge -> Regional VMS Hubs -> Central State Surveillance Cloud):

- Tier 1: Edge Ingestion & Inference (80,000 Cams)  
Local Junction Cameras (RTSP/ONVIF) -> Edge Micro-Gateways (NVIDIA Jetson Orin / Intel OpenVINO) for ANPR OCR, Bounding Box Inference, Motion & Loitering Flags, and RTSP-to-HLS Transcoding.
- Tier 2: Regional VMS & Aggregation Hubs (6 Zones)  
Regional Gateway Cluster with Kafka Partitioned Ingestion, 30-Day Video Recording Warm Storage, Local Watchlist Fast-Trie Matching (<5ms), and Local Officer Dispatch Node.
- Tier 3: Central Command & AI Intelligence Cloud (State HQ)  
Central Platform Cluster (Node.js/FastAPI) providing Global GIS Spatial Reconstruction, Long-Term Cold Storage, Cross-Zone Vehicle Re-ID, Central Watchlist Synchronization, and Admin Controls.

## 2. Video Bandwidth & Low-Bandwidth Optimization Strategy

Streaming 80,000 full raw 1080p RTSP video feeds to the cloud would require over 280 Gbps of sustained bandwidth. Our architecture solves this via:

1. Metadata-First Ingestion:  
Edge devices do NOT stream 24/7 continuous video to central cloud. Only lightweight JSON telemetry (plate, timestamp, GPS, confidence, bounding box) is streamed continuously (~2 KB/sec per camera = ~160 MB/sec for all 80,000 cameras).
2. On-Demand Video Retrieval (WebRTC / HLS Relay):  
Live video stream is only fetched and transcoded when an operator clicks 'Live View' or when a High-Priority Watchlist Alert is triggered.
3. Dynamic Resolution & Frame-Rate Throttling:  
Non-critical background cameras stream at 5 FPS / 720p. When motion or vehicle presence is flagged, frame rate ramps automatically to 30 FPS / 1080p.

## 3. Database Indexing & Scalability Strategy

- Relational Metadata (PostgreSQL / RDS Aurora):  
User profiles, role permissions, camera registry, watchlist entries, and alert incident audit logs. Indexed by (camera\_id, timestamp) and (identifier, active).
- Time-Series Telemetry (TimescaleDB / ClickHouse):  
Vehicle movement detections partitioned by day and zone\_id. Capable of indexing 100,000+ ANPR detections per second with sub-second vehicle route queries.
- In-Memory Caching (Redis Cluster):  
Watchlist fast lookup with in-memory Trie and Hash Sets (O(1) match complexity). Active WebSocket session routing and pub/sub distribution.

## 4. Hardware & GPU Accelerator Requirements

- Edge Compute: 4,000x NVIDIA Jetson Orin Nano (20 cameras/unit)

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- Regional Aggregation: 160 Regional Servers (32-Core, 64GB RAM) across 6 Zones
- Central Cloud Cluster: 8 Scaled Kubernetes Pods (32 vCPU, 128GB RAM)
- Central GPU Pool: 32 Cloud GPUs (NVIDIA A10G/L4) for Heavy Tactical Re-ID

## 5. Cybersecurity & Network Segmentation

- Zero-Trust Network Segmentation: Edge cameras communicate only within isolated VLANs via encrypted mTLS.
- Role-Based Access Control (RBAC): Fine-grained permissions (HQ Admin, Traffic Officer, RTO Inspector, CID Investigator).
- Tamper-Resistant Audit Trail: Every user login, camera onboarding, and alert acknowledgement is signed with timestamp and immutable audit log entry.